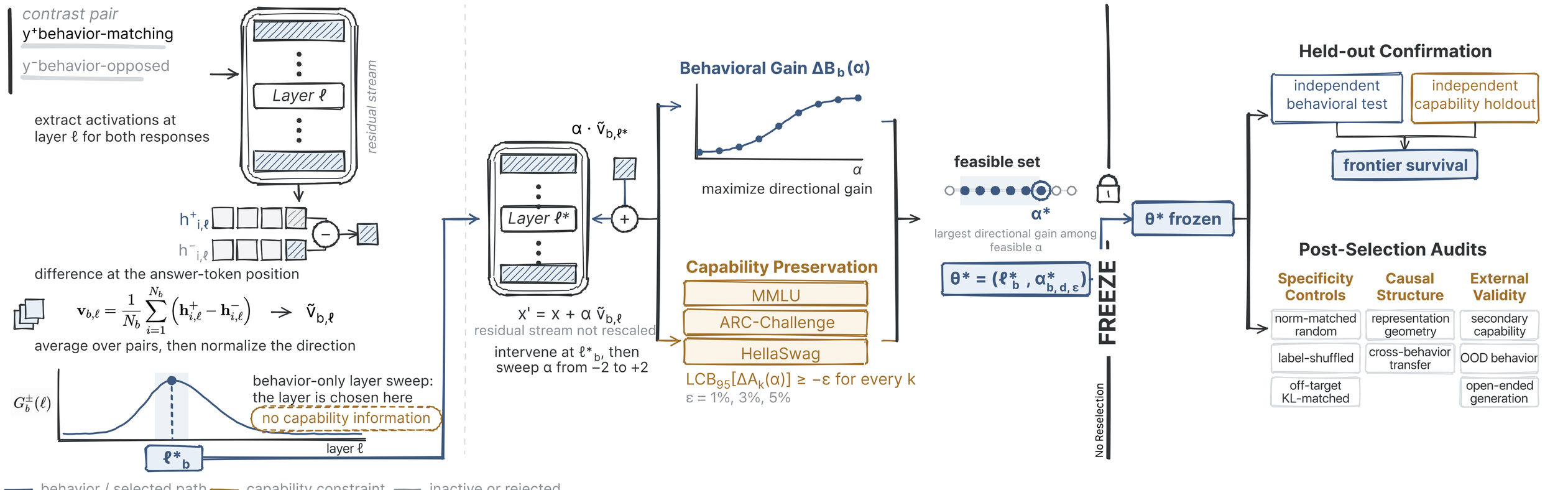

A. Behavior Direction & Layer Selection
contrast pair
y⁺behavior-matching
y⁻behavior-opposed
extract activations at
layer ℓ for both responses
Layer ℓ
residual stream
h⁺ i,ℓ
h⁻ i,ℓ
difference at the answer-token position
$\mathbf{v}_{b,\ell} = \frac{1}{N_b}\sum_{i=1}^{N_b}\left(\mathbf{h}^{+}_{i,\ell} - \mathbf{h}^{-}_{i,\ell}\right) \rightarrow \tilde{v}_{b,\ell}$
average over pairs, then normalize the direction
$G_b^{\pm}(\ell)$
behavior-only layer sweep:
the layer is chosen here
no capability information
layer ℓ
ℓ*b
B. Strongest Feasible Intervention
α · ṽb,ℓ*
Layer ℓ*
x' = x + α ṽb,ℓ
residual stream not rescaled
intervene at ℓ*b, then
sweep α from −2 to +2
Behavioral Gain ΔBb(α)
α
maximize directional gain
Capability Preservation
MMLU
ARC-Challenge
HellaSwag
LCB95[ΔAk(α)] ≥ −ε for every k
ε = 1%, 3%, 5%
feasible set
α*
largest directional gain among
feasible α
θ* = (ℓ*b , α*b, d, ε)
FREEZE
No Reselection
C. Confirm Frozen Operating Point
θ* frozen
Held-out Confirmation
independent
behavioral test
independent
capability holdout
frontier survival
Post-Selection Audits
Specificity
Controls
norm-matched
random
label-shuffled
off-target
KL-matched
Causal
Structure
representation
geometry
cross-behavior
transfer
External
Validity
secondary
capability
OOD behavior
open-ended
generation
behavior / selected path
capability constraint
inactive or rejected